

GPS Navigator

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Global positioning system (GPS) is a wonderful technology that has made navigation

systems highly accurate and efficient in recent years. All the mobile phones these days come with an in-built GPS receiver through which you can easily find out your current location anywhere on the earth. The only thing required is an unobstructed line-of-sight to four or more GPS satellites.

A GPS navigational device is any device that receives GPS signals and processes them to extract information for determining its exact location. Presented here is such a GPS device with a tracking record system. It shows the path traversed by you from the initial position, so you are never lost at unknown locations and can always come back to the initial point.

Circuit and working

Fig. 2 shows circuit of the GPS navigator. The circuit is built

TABLE I
NMEA Output Sentence

Option	Description
GPGLGA	Time, position and fix type data
GPGLSA	GPS receiver operating mode, active satellites used in the position solution, and DOP values
GPGLSV	Number of GPS satellites in view, satellite ID numbers, elevation, azimuth and SNR values
GPRMTC	Time, date, position, course and speed data. Recommended minimum navigation information
GPVTG	Course and speed information relative to the ground

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when S1 pressed
TP3	Train of pulses (when GPS module is connected)



Fig. 1: Author's prototype

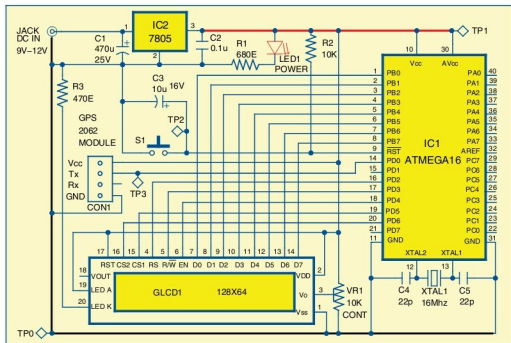


Fig. 2: Circuit of GPS Navigator

around microcontroller ATmega16 (IC1), 5V voltage regulator 7805 (IC2), GPS module (connected at CON1), graphical LCD (GLCD1) and a few other components.

The circuit is powered by a 9V/12V adaptor. Regulator IC2 provides 5V regulated supply for the circuit to operate. LED1 indicates presence of power in the circuit.

Microcontroller IC1 running at a clock frequency of 16 MHz communicates with the GPS receiver module via serial protocol. Tx pin of the GPS receiver is connected to Rx (PD0) pin of microcontroller IC1. The GPS receiver continuously transmits data at 1Hz update rate.

A 128×64-pixel, KS0108-controller-based GLCD is used to display